



Fig. 3: Obstruct shapes in ZenitPCB



Fig. 4: Renaming the components

text, shape, IEEE symbols and Field-Text could be added.

ZenitPCB GerberView

After creating Gerber and NCD drill files, the system creates an ASCII file bearing the name and the folder of gerber files. With the button 'Open GerberView,' the ZenitPCB GerberView program can be opened to see gerbers according to the ASCII file.

Other useful PCB design software

In a survey conducted by EFY, readers shared their views on other useful PCB design software:

EAGLE, *EAGLE*, aka, 'Easily Applicable Graphical Layout Editor,' is an easy-to-use, user-friendly layout editor software, used by a large number of PCB designers. It is developed

by CADSOFT and is available for Windows, Linux and Mac OS. Its auto function routes and places all the components automatically. It uses industry standards and is basically for schematics and board layout design only – no practical simulation. Although not a newbie-friendly, it is technically full-featured even in the free version.

One of the active members on the EFY Forum and Facebook page, Tapas Pattnaik, said, "Cadsoft EAGLE is the best PCB design software. Its schematic editor, auto-router and layout editor are awesome. The V6 version of Eagle is far better than others with its XML data structure. With the help of it we can easily transfer the data files from/to other CAD programs.

Pulsonix. With a totally integrated schematic capture and PCB design editor, Pulsonix is believed to be the best competitor to Eagle PCB design software. The only downside is that the free demo has 'save' option disabled and the cheapest full-featured version of the software is way too costly.

Proteus. Developed by Labcenter Electronics, it is used for PCB design as well as microprocessor simulation and schematic capture. It is an all-in-one product that allows schematics to board layout through to simulation on a high level. Being easy to use, it is good for hobbyists as well as students working on small projects.

PCB design steps with ZenitPCB

1. Place components in ZenitCapture
2. Route all components
3. Package parts
4. Create net list
5. Open ZenitPCB
6. Place board outline
7. Import net list
8. Route the circuit

What's New on ZenitPCB 1.6.5

- Library: Edit attribute, find footprints
- Layout: New mechanical layers and document layers
- Capture: Auto-rename, mirror part and move part without connections

System requirements

OS: Windows XP SP3 or higher version
CPU: Intel Atom D525 @ 1.8 GHz
RAM: 2GB DDR3
HDD: At least 50 MB free space

Pad2Pad. It is a simple-to-use free-ware tool that can be used to design, analyse, price, order and manufacture PCBs. Requiring few resources, it can work with minimum system requirements of Windows 2000/XP/Vista/7, Pentium 233MHz or higher, 128MB RAM and less than 30 MB of disk space. If you are not very familiar with this type of software, it would be a little difficult to achieve the desired board. It is also not easy to design a big PCB. An interesting feature is the capability to order your design once finished.

ExpressPCB. It is a free CAD software that is fairly simple with limited options. It is easy to learn, but you can use it only to order PCBs from the company. Also, it lacks some really useful features. There is no auto annotation option for the drawn circuit. You cannot choose the PCB footprints (packages) for components, to prepare your circuit for the PCB editor. Also, nets list for ExpressPCB or other PCB editor software cannot be generated.

KiCAD. It is an Open Source software suite for electronic design automation that can be run on FreeBSD, Linux, Microsoft Windows and Mac